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ChemProp multi-task models for predicting ADME properties in the Polaris challenge

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Abstract

Accurate prediction of ADME (Absorption, Distribution, Metabolism, and Excretion) properties is a key challenge in drug discovery. In the Polaris Antiviral ADME Prediction Challenge, we developed and benchmarked multi-task directed message passing neural network (D-MPNN) models using ChemProp, trained exclusively on a curated collection of public datasets comprising over 55 tasks. We demonstrate that high-quality data curation, coupled with multi-task learning, worked well in the context of prediction of the ADME endpoints in this challenge. Our final models incorporated both experimental assay data and calculated properties. Using only public data, our approach achieved second place among 39 participants, surpassed only by a model using proprietary data. These results highlight the power of combining well-curated data together with multi-task ChemProp models for robust ADME prediction and set a benchmark for future community-driven efforts. To support continued progress, we publicly release the dataset and data processing scripts used in this work.

Keywords

: D-MPNN, multi-task machine learning, ADME prediction

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Abstract

Accurate prediction of ADME (Absorption, Distribution, Metabolism, and Excretion) properties is a key challenge in drug discovery. In the Polaris Antiviral ADME Prediction Challenge, we developed and benchmarked multi-task directed message passing neural network (D-MPNN) models using ChemProp, trained exclusively on a curated collection of public datasets comprising over 55 tasks. We demonstrate that high-quality data curation, coupled with multi-task learning, worked well in the context of prediction of the ADME endpoints in this challenge. Our final models incorporated both experimental assay data and calculated properties. Using only public data, our approach achieved second place among 39 participants, surpassed only by a model using proprietary data. These results highlight the power of combining well-curated data together with multi-task ChemProp models for robust ADME prediction and set a benchmark for future community-driven efforts. To support continued progress, we publicly release the dataset and data processing scripts used in this work.

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Introduction

Accurate prediction of absorption, distribution, metabolism, and excretion (ADME) properties is crucial for drug discovery and development. Computational models, particularly machine learning (ML)-based approaches, have gained increased interest in industrial applications, due to their ability to infer complex structure-property relationships from large datasets. The use of large-scale multi-task models for ADME prediction has already demonstrated its value,¹⁻⁵ especially in the form of directed message-passing neural networks (D-MPNN's) like ChemProp multi-task models,⁶⁻⁸ which take advantage of shared information across multiple ADME endpoints to enhance predictive accuracy.^{4,9} Multi-task learning has been shown to improve model generalizability by leveraging correlated tasks and mitigating the impact of data sparsity.¹⁰⁻¹³

However, the development of robust and generalizable ML models for ADME prediction remains an ongoing challenge due to the limited availability of high-quality experimental data and the inherent complexity of biological systems, in addition to the challenges associated with data quality and curation, especially in the public domain.⁴

Moreover, reports of performance of machine learning based methods in literature typically overestimate the true performance of benchmarked models,^{14,15} as historically they were based on random splitting of training and test sets, which introduces the risk of data leakage, i.e. in this context, the presence of molecules with high structural similarity between sets. Time- and scaffold-based splits could provide a more realistic assessment of the expected prospective performance,^{2,6} but can also be overoptimistic, therefore they should not be considered in isolation either.¹⁶ Furthermore when training large scale multi-task models, data (tasks/assays) should be balanced across folds.¹⁷ This can be achieved by a sequential approach of clustering followed by data balancing through linear programming.¹⁸

Data curation is another key challenge in the field. If not handled appropriately, there is a significant risk of issues like overfitting and data leakage, which can further contribute to overly optimistic benchmarking performance.^{19,20} One example that illustrates this issue is the AstraZeneca deposited sets for clearance in hepatocytes,²¹ which was curated by the TDC,²² without taking the different species into account for that specific task (see supplementary Table S1). Nevertheless, such community-wide availability of ML-ready resources is invaluable to progress in the field. These datasets make it possible to evaluate a wide range of computational tools and methodologies across different groups, to enable more objective performance assessment and benchmarking. One of these community-driven effort has been pursued by Polaris with the ambition of hosting ML-ready datasets.

Recently, they hosted their first competition for the prediction of: 1) ADME properties 2) protein-ligand binding poses and 3) binding affinities. Data was mainly taken from the COVID Moonshot initiative, which was a collaborative open science effort for the discovery of novel SARS-CoV-2 3C-like protease inhibitors.²³

In this study, we explore various ML models to predict ADME properties as part of the Polaris antiviral ADMET prediction challenge. Our approach consisted of:

- 1) A high-quality curated dataset consisting of publicly available datasets, containing over 55 individual tasks.
- 2) Benchmarking and optimization of various single- and multi-task models such as Random forests, XGBoost, PyBoost and ChemProp^{6,7,24,25}
- 3) The use of predicted descriptors/properties as tasks, as previously done in the group for the SAMPL7 challenge.²⁶

The final models that were used for the submissions were ChemProp models trained on public data and several predicted properties (see methods for further details). With these models in

hands we ranked 2/39 in the final leaderboard, only surpassed by models created with proprietary methods and data. This achievement highlights that properly curated public data can already provide high quality and impactful models, for this reason we are releasing the dataset and curation scripts that were used so that they can be of use to the community.

Methods

Datasets and data curation

Data was fetched from different sources, including:

- 1) ChEMBL^{27,28}
- 2) Opera^{29,30}
- 3) TDC²²
- 4) Individual sources: Biogen,³¹ AstraZeneca,²¹ Martel et al.,¹⁴ Lombardo et al.³² and finally METLIN SMRT.^{33,34}

Datasets were downloaded, subsequently different standardization and curation functions were applied:

F3_csv_unpivoted_to_standard_transformed_curated – which curates the chemistry and then transform the to-be modelled y_values, per dataset/assay/task. For the chemistry curation we used a procedure conceptually similar to the, and based on the ChEMBL standardizer.³⁵

In short:

- a) Molecules are generated from (CX)smiles
- b) Heterolytic bonds are split matching the following SMARTS pattern: [Li,Na,K;+0]-[#7,#8;+0]"
- c) Mixtures are removed
- d) Non-organic molecules are removed

- e) Molecules are canonicalized, also focusing on chirality (enhanced stereo flags, meso chirality, single wedges and further standardization of racemic groups)
- f) Molecules are converted back to (CX)smiles
- g) Apply corresponding transformation to the y_values (e.g. log, logit, no transform).

F4_csv_unpivoted_std_transf_cur_to_averaged – a function to average the dataset of interest based on the output smiles of F3, while taking censors into account

F5_csv_unpivoted_std_avg_append_and_pivot – a function to take a list of output datasets from F4 and pivot them together into one multi-task ML-ready dataset. These scripts are shared on github: <https://github.com/lenselinkbart/Datautils>

Applying these functions led to a curated dataset, an overview of the number of datapoints per task in this dataset can be found in *supplementary table 2*. This dataset and final models have been shared and can be downloaded from: <https://zenodo.org/records/15602150>

Train, test, validation set construction

We constructed two different types of test sets, with different goals. a) A 5-fold balanced, chemically diverse split (ML_subset)¹⁸ – used for hyperparameter optimization in ChemProp, unless mentioned otherwise. Alternatively, a 5-fold random split for the single-task tree-based models, using only the molecules with values for the respective modeled properties. b) A diverse holdout-test set, similar to the challenge test set data – used for deciding which models to use in the submission – obtained by selecting training set molecules with a Tanimoto similarity >0.6 and subsequent clustering, selecting a total of 26 molecules. This holdout test set was used for the majority of benchmarks, including for the tree based models. For final ChemProp model training (interim/final submission to the challenge), models were trained on the full dataset, as mentioned below.

D-MPNN training

We used an internal fork of ChemProp 1.5.1, with the following, minor modifications:

1. Implementation of an evidential_bounded loss function: this loss function is a combination of the bounded_mse and evidential loss function, setting the evidential loss to zero if the censor threshold value is met.
2. A ratio between layers (controlled by ffn_ratio and mpn_ratio), allowing control of the size of different layers (e.g. layer_2 size = layer_1 size * ratio)
3. New activation/aggregation functions, like for instance VPA.³⁶
4. Modifications so pretraining can be done when the number of tasks is different, if this is the case the output layer is replaced by a new output layer. This strategy was used for some of the interim submission models, pretraining on the public dataset, and fine-tuning on the Polaris data (not further discussed in this manuscript).
5. Several helper and integration scripts, using MLflow,³⁷ Optuna³⁸ and several other MLOps scripts (e.g. API/infrastructure integration).

Unless specified otherwise we trained an ensemble of 10 ChemProp models, using different seeds for the data splitting, and using 10% of the data as a random validation set (for determining the best epoch). For the models that included calculated properties, properties were calculated for *all* Polaris challenge molecules, including leaderboard test set molecules. These properties were used in the forward property addition benchmark, where we iteratively built models adding one property at a time – keeping the best performing property and removing the worst performing one. For the final submission we used a weighting scheme, based on the total evidential uncertainty. We tested weighting inversely by the uncertainty but also taking the average of the top n – most certain predictions (t2 = top 2). The latter was used for some of the final models.

Finally, for the submission (interim/final) models the same procedure was followed but the internal hold out test set (26 molecules) was also added to the training dataset.

XGBoost, PyBoost and Random Forest model training

For all three approaches, we trained a single-task model for the individual properties. The models were set up similarly to the methodology reported previously by Fang et al.³¹ As a molecular representation, we used 305 2D RDKit descriptors in addition to a 1024-bit Morgan fingerprint with radius 2. Production models were created by performing a random split on the available data for each property in combination with the selected hyperparameters (see below for the hyperparameter tuning of each approach). The internal hold out test set was excluded, and 90% of the remaining data was used for training, with the remaining 10% used as the validation set. The models were then evaluated against the hold out test set.

XGBoost hyperparameter tuning

For each of the properties, we performed a hyperparameter tuning with five individual search grids and combined the best performing parameters from the five-fold cross-validation in the production model.

PyBoost hyperparameter tuning

Instead of doing hyperparameter tuning, we adapted the ideal parameters identified for the XGBoost model and used default settings otherwise in the production model.

Random Forest hyperparameter tuning

For each of the properties, we performed a hyperparameter tuning using a grid search. The best performing parameters were used in the production model.

ESOL calculations

We calculated the solvation free energy using the Schrödinger suite.³⁹ Molecules were prepared using Ligprep, default settings, except for epikx (-pht 1).⁴⁰ Subsequently the ESOL Jaguar workflow was used to calculate different QM properties. Recommended settings were used (n_conformers_conf_search = 12, n_conformers_to_score = 4).⁴¹ The output of this workflow was solvation and hydration free energy, Esol, HOMO, and LUMO. The HOMO/LUMO Gap was calculated based on this (HL gap = HOMO-LUMO). This 3D output was mapped back to the 2D structures by taking the average values, this was done for pragmatic reasons, and because for most compounds (508/530) the standard deviation was <1kcal/mol (for Esol).

Other software

We calculated other properties, related to polarity (PSA/ 3D-SAPSA) or lipophilicity (logP/logD). Like previously done in the SAMPL7 challenge,²⁶ we used the ADMET predictor from Simulations Plus, to calculate logP and logD.⁴² logD was calculated at different pH values: pH – 4,6,7.4,8 and 10.

RDKit was used to calculate TPSA,⁴³ additionally we calculated the 3D Solvent Accessible Polar Surface Area (3D-SAPSA) using RDKit for the generation of conformers and the partial charges were originated from aimnet2⁴⁴ or espaloma.⁴⁵ The implementation of FreeSASA in RDKit was used for the SA calculation itself. Polar atoms were identified as nitrogen and oxygen atoms, hydrogens bonded to nitrogen or oxygen, and any atom exceeding a specified partial charge cutoff (0.6). The solvent-accessible surface area was computed with a standard probe radius of 1.4 Å. The calculation then output either the average 3D-SAPSA for all conformers or the minimum value observed for these conformers.^{46,47} The code can be found on github: https://github.com/lenselinkbart/3d_sapsa

Results

Baseline model

Based on the multi-task dataset we did a comparison between single and multi-task models. This comparison was done on the 5-fold chemically diverse split, as shown in Figure 1. We trained models using default settings in ChemProp; in this case single-task models were trained. Perhaps not unexpected, the multi-task model outperformed the single-task D-MPNN models for every property. The biggest improvement in performance going from single- to multi-task was seen for logD, which could be attributed to the fact that our ML-ready dataset contains many assays/records related to lipophilicity (e.g. see supplementary Table S2).

The performance of the baseline multi-task model was already competitive for the hold-out test set, as shown in Figure 2. In fact this model was used for interim submissions 1 (for logD, HLM and MLM) and 2 (for logD), demonstrating that at least for some properties, the data – and its careful curation – matters more than the (hyper)parameters.

Tree-based models underperform

The performance of the tree-based models we tested is summarized in Figure 3. There is no indication that either XGBoost or PyBoost is the better model across the five properties. Both perform similarly in predicting LogD, KSOL and MDR1-MDCKII, while XGBoost outperforms PyBoost for HLM prediction and PyBoost outperforms XGBoost for MLM prediction. RF also performs similarly, with the exception of LogD prediction, where it underperforms on this hold-out test set. When comparing the tree-based models to the baseline ChemProp multi-task model, the ChemProp model clearly outperforms both XGBoost and PyBoost in most properties, with the exception being the MDR1-MDCKII permeability assay. As the tree-based models were constructed for a single-task output, whereas the reference

ChemProp model as a multi-task model, this could indicate that in the case of MDR1-MDCKII, the performance of the model might be compromised by molecules in the training set that are similar to molecules in the test set, but have associated tasks which are uncorrelated to the MDR1-MDCKII readout.⁴⁸

Hyperparameter optimization has a (small) positive impact for some properties on the hold-out test set

After training several baseline, multi-task models we performed a hyperparameter optimization run, using Optuna.³⁸ Our overall strategy was to optimize parameters on the 5 fold chemically diverse split, and afterwards decide which final models to use in the submission based on the internal, hold out test set. 100 iterations of hyperparameters optimization were done on the 5-fold chemically diverse split, selecting the following parameters: `ffn_hidden_size` = 2000, `depth` = 2, `dropout` = 0.173, `ffn_num_layers` = 3, `ffn_ratio` = 0.9, `activation` = "ReLU", `max_lr` = 0.0005, `epochs` = 50, `bias` = False, `init_lr` = 0.0001, `final_lr` = 0.0001, `aggregation` = "mean". Performance was assessed on the hold-out test set, by assessing the MAE (metric used in the challenge), as shown in Figure 4. Interestingly for both logD and KSOL, performance slightly decreased while for the other properties (MDCK, HLM, MLM) performance increased (albeit not significantly). The biggest gain in performance was observed for MDCK, for which the MAE dropped from 0.40 to 0.29 – suggesting that this task benefited the most from a more complex model (i.e. a model with more parameters).

Forward property addition further improves the model

As a final step we aimed to add calculated properties to the model; due to time constraints, this optimization was done on the hold-out test set (running the risk of overoptimization). Properties were added, one at a time, keeping the ones with the best performance and deleting the ones with the worst performance. For the first iteration of models, we trained both a default and an optimized model, and took forward the one with the best performance. An overview of which models were trained can be found in Table 1, alongside with which properties were iteratively added. Several properties had a positive impact on model performance, the best improvement after the first iteration was observed upon adding the HL_gap to MLM (RMSE: from 0.368 to 0.323) – interestingly this property was selected also for other assays (HLM & MDCK). Indeed it has been shown recently that QM properties can be used to improve predictive models for ADME.⁴⁹ Unfortunately, due to time constraints, we were not able to use this model for MLM in the final submission; we used instead the model that was used for the intermediate leaderboard. This model was an optimized one, but not including calculated properties.

Final leaderboard submission and results

After training the models with calculated properties added, we validated on the benchmark set if a weighting scheme, based on the uncertainty, improved the results. For logD and KSOL this was the case, therefore we decided to use the average of the two/three most certain predictions respectively.

Comparing the performance on the final leaderboard (Table 2), and the one on the hold-out test set, it became apparent that, in hindsight, optimizing on the hold-out test set (e.g. the forward calculated property addition) was perhaps not the best strategy. For most properties this strategy worked, except for LogD, where our model underperformed, with a MAE of 0.323 versus 0.253 of the number one model in the challenge (ranking us number 5 for LogD). If we had used our initial model, i.e. an unoptimized “base” multi-task one, we would have ranked second with a MAE of 0.262. Further work is needed to determine a better strategy for selecting the optimal number and combination of tasks.

Discussion and outlook

In this manuscript, we have discussed our approach in the Polaris challenge. Key to this approach was using multi-task models trained on a reasonably sized collection of public, curated datasets. In total 55 different tasks were used (see supplementary Table S2) -with a strong bias towards lipophilicity datasets. Moving forward, more data on non-lipophilicity endpoints would be of value. Initiatives like Polaris, openADMET, TDC, ncats ADME, Opera, among others, and publications are therefore of great value to the community to enable researchers to build better quality models.^{21,29,31,50}

Still given the vastness of the chemical space,⁵¹ (in)compatibility/heterogeneity of different assays and relatively small number of available datapoints, more research and open data would be needed. In our opinion multi-task models are part of the solution for this, given that they can benefit from implicit correlations between assays, and help to avoid having to harmonize/scale highly correlated but nominally different assay results, even when the dataset is sparse. For this purpose, ChemProp/D-MPNN seems to be the library of choice; although Fang et al.³¹ have shown that they can build better models using XGBoost, multi-task models were not considered. However, in the context of kinases, but also for lipophilicity and in this manuscript

(e.g. Figure 1), we have clearly demonstrated that single-task models underperform. Moreover, for ADME prediction, multi-task models have been the model of choice in many pharmaceutical companies.¹⁻⁵ Nevertheless, as deep learning-based models are generally assumed to outperform traditional tree-based methods only when the training set becomes sufficiently large, it is unclear if the superior performance of the ChemProp model can be attributed to the model itself, to the molecular representation, or to the multi-task aspect.

As another potential solution to the overall sparsity of ADME data, calculated properties can be considered, especially when they are correlated with assays of interest. Lawrenz et al.⁴¹ have shown that solvation free energy is correlated with $K_{p,u,u}$, but also weakly correlated with permeability in MDCK. For that reason, this ESOL workflow was adopted here. As an additional output, both the HOMO and LUMO were calculated, from which we derived the HOMO-LUMO gap. Recently Fallani et al.⁵² has shown that (pre)training models on QM calculated properties can substantially increase model performance; here we observed something similar, similarly here, the properties were used as tasks and not as features/descriptors. PSA was calculated based on both the 2D structure and the 3D structure. This was done because Moblitz et al.⁵³ have shown that 3D PSA can be correlated with permeability. Moreover, to account for ionization, we tested adding logD calculated at different pH values by Simulations plus. As another option COSMO-RS could be considered to improve logP/logD predictions, given its excellent performance in the SAMPL6 challenge.⁵⁴ In general, the overall strategy is to augment 2D machine learning models with 3D and physics based approaches, potentially offering various kinds of properties that could be calculated.

As a final note we would like to stress the importance of data curation. We think researchers in the field are spending comparatively too much time on the algorithm side, versus data curation and standardization. Here we have shown that good data + good algorithms beats

algorithms that have claimed superiority (e.g. the several papers claiming to have demonstrated superiority over ChemProp). Having standardized datasets available to the community, as mentioned above, through initiatives like Polaris, will help to further help the field to evolve.

Conclusions

In this work we have described our approach in the Polaris ADME challenge. Our submission ranked overall second, and even first for MLM, while only using public data. For this, several factors were key: 1) Meticulous data curation and standardization; 2) being able to construct meaningful benchmark sets; 3) ChemProp/D-MPNN's architecture 4) adding calculated properties (for most tasks); 5) multi-task models. We anticipate that this manuscript will provide some interesting insights on how performant models can be constructed for ADME even with only public data available. We anticipate that adding proprietary data will only improve performance and/or expand the applicability domain. By releasing the dataset (including calculated properties) & scripts that were used for this challenge to the community, we hope that it can be further used and analyzed. We believe it can serve as a solid building block for the foundation of open-source ADME models.

Declarations

Funding

Not applicable

Conflicts of interest/Competing interests

Not applicable

Availability of data and material

The data used in training the final model, and final models will be shared on: ...

Code availability (software application or custom code)

The code to do the data curation has been shared:

Acknowledgments:

Not applicable

Authors' contributions

KG/GT/EBL: involved in curating the data & performing the experiments and the analyses

KG/EBL/JH/SDC defined the scope and strategy of the work and wrote the manuscript.

Figures

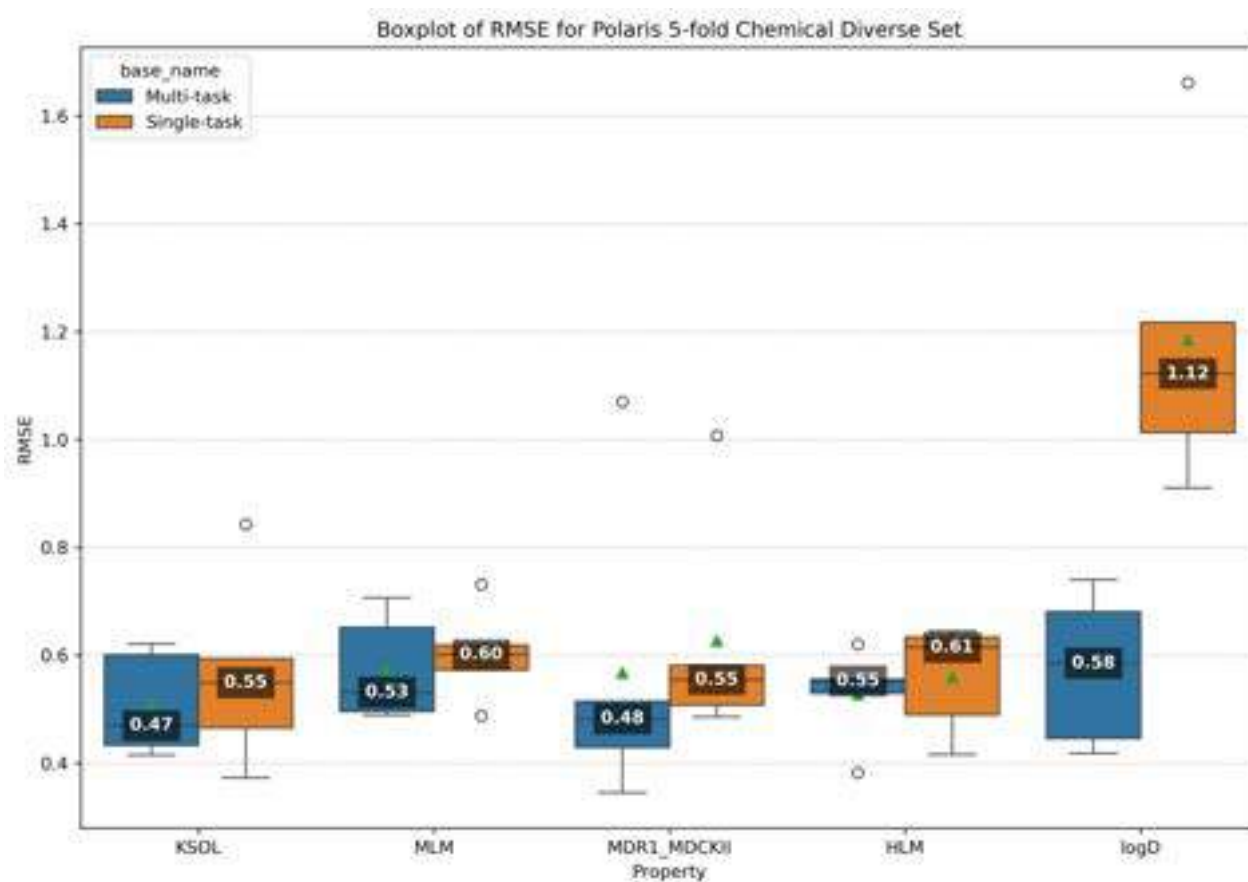


Figure 1. Boxplots of the RMSE of single- versus multi-task ChemProp models on the 5 fold chemical diverse splits. The values shown in the boxes, represent the median RMSE (also indicated by the line)

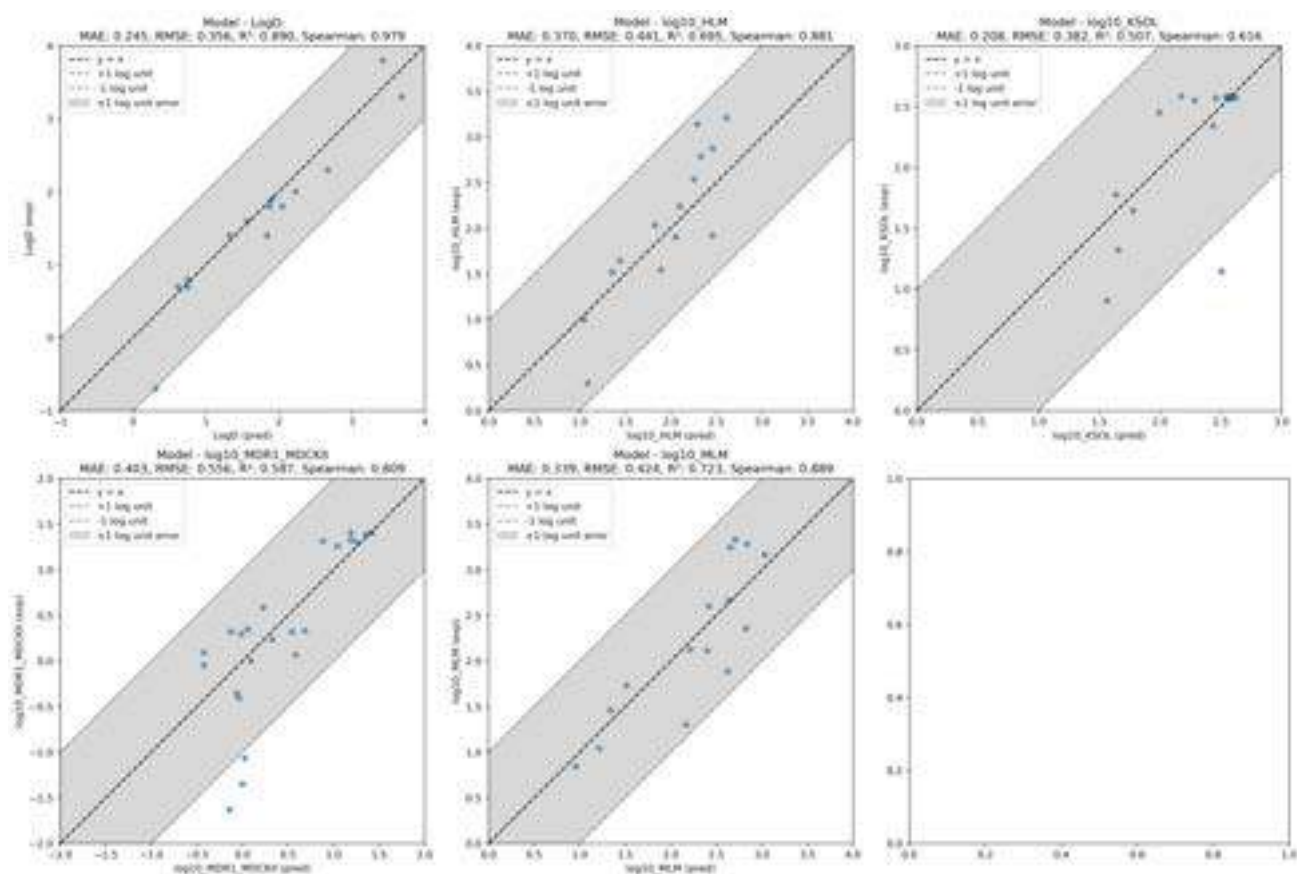


Figure 2. Scatterplot of the predictions of the multi-task ChemProp models on the internal hold-out test set versus the experimental values in the log10 scale. MAE, RMSE, R^2 , and spearman rho were calculated per assay and are shown.

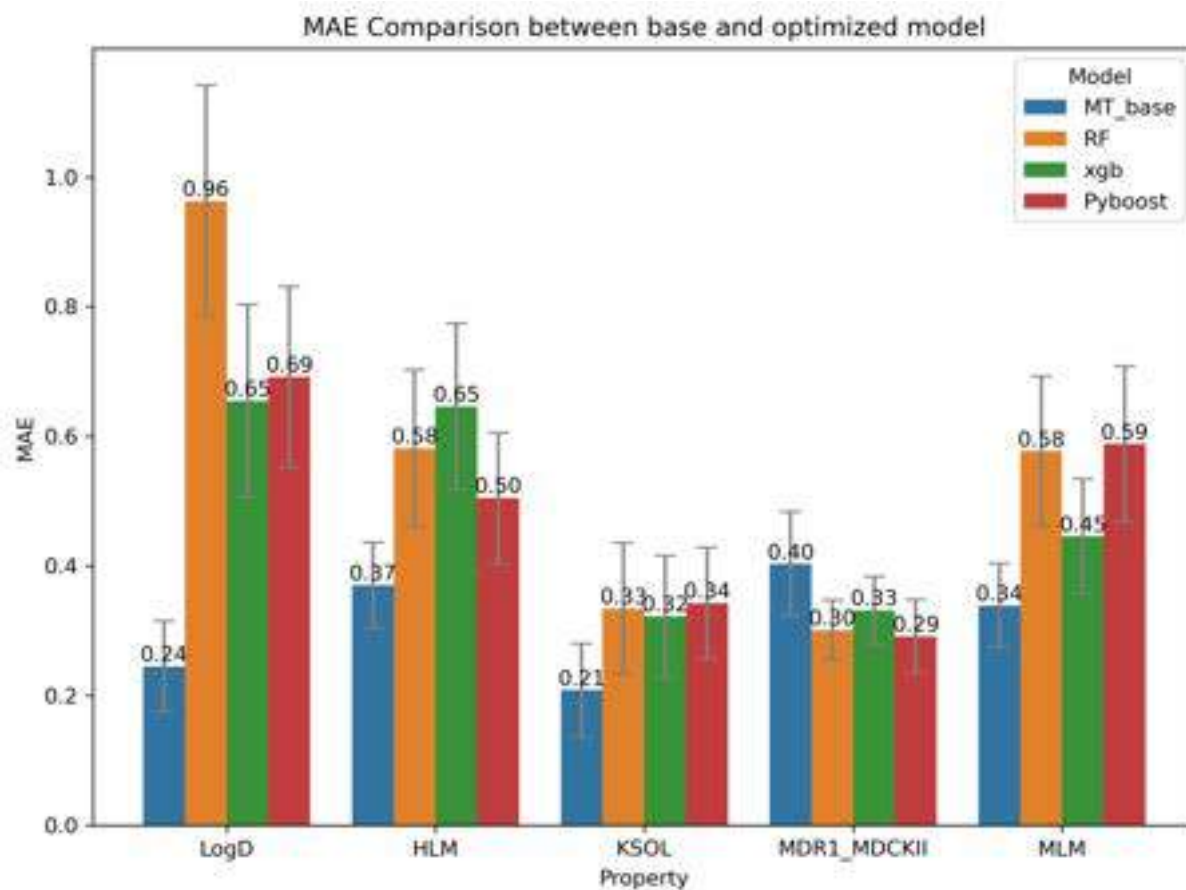


Figure 3. Bar chart comparison of the MAE on the hold-out test set for the Random Forest, XGBoost and PyBoost models compared to the baseline model (MT_base), for the 5 different properties. MAE's are shown above the bar. Standard errors are shown in grey, calculated using bootstrapping in mlxtend⁵⁵

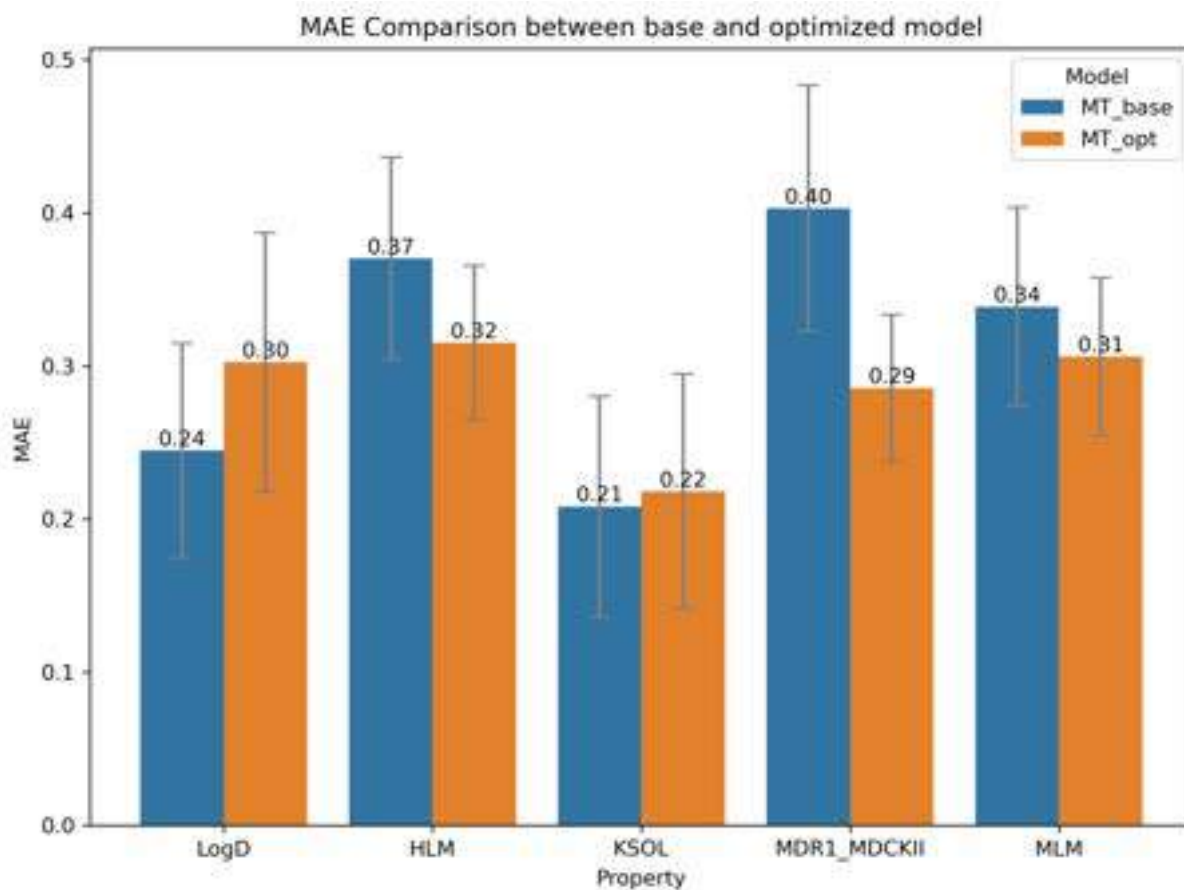


Figure 4. Barchart comparison of the MAE on the hold-out test set for the baseline model (MT_base) and hyperparameter optimized model (MT_opt), for the 5 different properties. MAE's are shown above the bar. Standard errors are shown in grey, calculated using bootstrapping in mlxtend⁵⁵

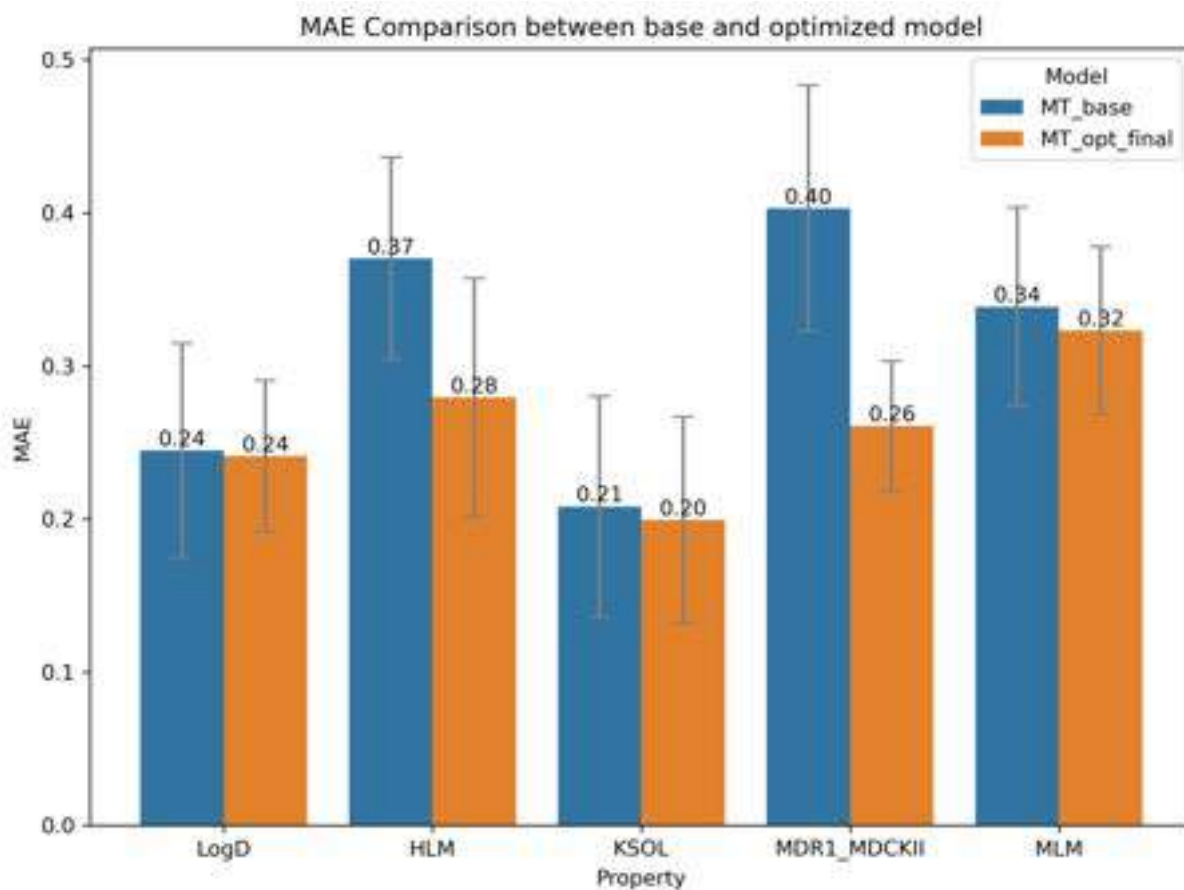


Figure 5. Barchart comparison of the MAE on the hold-out test set for the baseline model (MT_base) and the final used, model (MT_opt), for the 5 different properties. MAE's are shown above the bar. Of note for MLM we used a different model for the final submission as mentioned in the main text.

Tables

Table 1. An overview of the hyperparameters used (default or optimized), properties that were added and weighting scheme.

Model	Hyperparameters	Properties added	Weighting scheme ^a
LogD	Default	min+mean 3D SAPSA es / Splus_logD_pH7.4+10 / solvation free energy	T2
HLM	Optimized	HL_gap / min 3D SAPSA ani / Splus logD_pH4+7.4	none
KSOL	Default	TPSA / Splus logD_pH7.4 / mean 3D SAPSA es/ solvation free energy	T3
MDCK	Optimized	hydration free energy/ LUMO+ HOMO+ HL_GAP/ Splus logD pH4/ 7.4	none
MLM	Optimized	HL_gap / solvation free energy	none

^aWeighting was based on the total evidential uncertainty, taking the top n most certain predictions (i.e. T2 = average of two most certain predictions)

Table 2. An overview of the performance of the different models on the challenge test set.

Model	Performance (MAE) / Rank	Performance winner (MAE) / Submission
LogD	0.323 ± 0.021 / 5	0.253 ± 0.021 / prairiewarbler10
HLM	0.303 ± 0.022 / 2	0.275 ± 0.019 / prairiewarbler10
KSOL	0.112 ± 0.021 / 3	0.101 ± 0.022 / prairiewarbler10
MDCK	0.161 ± 0.013 / 2	0.151 ± 0.011 / prairiewarbler10
MLM	0.317 ± 0.024 / 1	<i>Second:</i> 0.338 ± 0.024 / prairiewarbler10

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Supporting Information

Table S1. Overview of the clearance data deposited in ChEMBL by Astrazeneca in comparison with the TDC.

Assay ChEMBL ID	Description	Frequency ChEMBL	Frequency TDC
CHEMBL3301370	Intrinsic clearance measured in human liver microsomes	1102	1102
CHEMBL3301371	Intrinsic clearance measured in rat hepatocytes	837	-
CHEMBL3301372	Intrinsic clearance measured in human hepatocytes	408	-
'71 + '72	1,020 “drugs” for hepatocyte clearance.	(1020 deduplicated)	1213 / 1020 deduplicated

Table S2. Overview of the number of datapoints per task, included in the curated dataset.

Property	Frequency
log10_SMRT_RT	79935
Opera_LogP	17896
TDC_LogAqSol	8756
Opera_Caco2_LogPapp	4503
ChEMBL_LogD_CHEMBL3301363	4196
Opera_human_Log_fup	3225
Biogen_Log_human_microsomal_CLint	3087
Biogen_Log_rat_microsomal_CLint	3054
Biogen_Log_MDCK_Efflux_Ratio	2642
Biogen_Log_Solubility_pH_6_8	2173

log10_Obach_2018_human_IV_PK_CL	1298
log10_Obach_2018_human_IV_PK_Vss	1266
Opera_human_hepatocyte_Log_CLint	1137
TDC_Caco2_LogPapp	897
log10_ChEMBL_CLint_CHEMBL3301370	744
Martel_LogP	706
log10_ChEMBL_CLint_CHEMBL3301371	595
log10_GLPG_Polaris_MDR1_MDCKII	394
ChEMBL_LogD_CHEMBL3431936	360
log10_GLPG_Polaris_KSOL	337
GLPG_Polaris_LogD	322
log10_ChEMBL_CLint_CHEMBL3301372	289
log10_GLPG_Polaris_MLM	271
log10_GLPG_Polaris_HLM	270
Biogen_human_Log_PPB_perc_unb	194
Biogen_rat_Log_PPB_perc_unb	168
ChEMBL_LogP_CHEMBL2098498	148
ChEMBL_LogD_CHEMBL2098497	148
log10_ChEMBL_CLint_CHEMBL1862765	141
ChEMBL_LogD_CHEMBL4384702	98
log10_Opera_RT	97
ChEMBL_LogP_CHEMBL633738	95
ChEMBL_LogP_CHEMBL4380721	84
ChEMBL_LogP_CHEMBL635201	80
ChEMBL_LogP_CHEMBL634672	74
log10_ChEMBL_CLint_CHEMBL5053348	71
log10_ChEMBL_CLint_CHEMBL4706205	65

log10_ChEMBL_CLint_CHEMBL4706204	65
ChEMBL_LogP_CHEMBL634679	59
log10_ChEMBL_CLint_CHEMBL2186143	57
ChEMBL_LogP_CHEMBL876041	57
log10_ChEMBL_CLint_CHEMBL2169662	57
log10_ChEMBL_CLint_CHEMBL3091165	53
log10_ChEMBL_CLint_CHEMBL1924275	53
log10_ChEMBL_CLint_CHEMBL1169305	51
log10_ChEMBL_CLint_CHEMBL2013638	51
log10_ChEMBL_CLint_CHEMBL2186144	51
log10_ChEMBL_CLint_CHEMBL3829797	51
log10_ChEMBL_CLint_CHEMBL3829798	51
log10_ChEMBL_CLint_CHEMBL1001484	50
log10_ChEMBL_CLint_CHEMBL3803439	48
log10_ChEMBL_CLint_CHEMBL3803438	47
log10_ChEMBL_CLint_CHEMBL3803440	46
log10_ChEMBL_CLint_CHEMBL2168486	45
log10_ChEMBL_CLint_CHEMBL4834037	42
log10_ChEMBL_CLint_CHEMBL5053351	41
log10_ChEMBL_CLint_CHEMBL3865206	40
log10_ChEMBL_CLint_CHEMBL4810045	37
log10_ChEMBL_CLint_CHEMBL3788415	32
log10_ChEMBL_CLint_CHEMBL2168487	32
